

Sepsis & Septic Shock - Summary

Key Symptoms: hypotension (SBP $\leq$ 100mmHg), altered mental status, increased respiratory rate, fever

Therapeutic Approach: BAPA

1. Breathing
2. Access
3. Perfusion
4. Antibiotics

Non-Pharmacological Treatment: Supplemental oxygen, fluids, mechanical ventilation, etc.

- **IV fluids:** 30 mg/kg
 - Crystalloids (e.g., normal saline or Ringers' lactate) – treatment of choice
 - Colloids (e.g., albumin solutions)

Pharmacological Treatment:

- **Vasopressors:**
 - Norepinephrine (first choice), epinephrine, dopamine, phenylephrine, vasopressin (adjuvant therapy, NOT monotherapy)
- **Others:**
 - Dobutamine, blood transfusions (when Hb $<$ 70g/L, etc.), corticosteroids (not routinely recommended, adjuvant ONLY)
- **Antibiotics:** broad-spectrum empiric antibiotics until cultures and sensitivities available
 - IV beta-lactams, extended-spectrum cephalosporins, carbapenems, fluoroquinolones
 - Choose based on source of original infection
 - +/- other agents for specific risk factors (Psa, VRE, MRSA) +/- anti-fungals
 - Narrow according to cultures and susceptibility

Additional Key Points:

- Take cultures before starting empiric treatment!!!
- Mortality increases each hour antibiotic therapy is delayed
- Introduce narrow/targeted antibiotic therapy once culture results are available
- Preventative measures while in the ICU
 - Glucose control, VTE prophylaxis, Stress ulcer prophylaxis, Impaired GI motility prophylaxis, Adequate caloric intake, etc.